01 PROBLEM AND RELATED WORK

SpikeDistill-RUL

Remaining useful life estimation for simulated turbofan engines

Research question. Can an attention-based recurrent teacher improve a compact spiking student's prediction of an engine's remaining cycles by transferring more than its final output?

Predictive maintenance needs a usable estimate of the time to failure from incomplete sensor histories. Here, a regression model predicts remaining useful life (RUL) in engine cycles from the latest 30 observed cycles. The project compares 12 baseline entries and four learnable spiking variants on NASA C-MAPSS FD001-FD004, then isolates hidden-state and attention matching in fresh controlled runs.

Main result. SpikeDistill-RUL improves raw test RMSE over the fixed LIF baseline on all four subsets. It uses 88.87%-88.97% fewer inference parameters than its teacher. A conventional model achieves the lowest mean RMSE on every subset.

Literature review

Prognostics and sequence models. Saxena et al. [1] use simulated gas-path deterioration to produce run-to-failure trajectories. The benchmark supports controlled comparisons, while transfer to field engines remains untested. Zheng et al. [2] apply LSTMs to RUL estimation. LSTM memory [3] and GRU gates [4] motivate two alternatives to Elman-style recurrence [16]. The bidirectional design of Schuster and Paliwal [5] motivates the two-pass window encoder. Shah et al. [6] combine attention and bidirectionality in a sequence-to-sequence prognostic system. Our teacher uses a scalar readout after pooling, with both passes restricted to the observed window. Li et al. [17] demonstrate convolutional RUL features, motivating the local filter in the CNN-LSTM control.

Spiking models and conversion. Neftci et al. [7] and Eshraghian et al. [8] describe surrogate gradients for hard spike thresholds. DIET-SNN [9] motivates learning leak and threshold from data with direct analog input. Rueckauer et al. [10] explain how rate-coded spikes approximate ReLU activations, motivating the conversion control. Their image-classification results leave open how well a short spiking simulation preserves scalar RUL calibration.

Teacher-to-student transfer. Hinton et al. [11] motivate teacher-output matching. We match scalar RUL by MSE. FitNets [12] introduces an intermediate hint and projection. We apply that idea per observed cycle, mapping student membrane states into the teacher's hidden space, and separately match temporal attention. The two loss removals test their individual contribution.

Explanation. Integrated Gradients [13] motivates path attribution. SHAP [14] and TimeSHAP [15] motivate perturbation analysis across sensor histories. We sample Shapley values over sensor-by-time blocks and complement them with attention, fixed-attention spike attribution and held-out sensor permutation.

Research gap. Output matching alone leaves the student's internal timing unconstrained. The project tests whether transferring state and timing information improves RUL estimates under a small spiking inference architecture.

02 DATA AND EVALUATION

NASA C-MAPSS protocol

One official test prediction per engine, with a fixed engine-disjoint validation split.

Each simulated row contains engine ID, cycle number, three operating settings and 21 sensor readings [1]. Training trajectories run to failure. Test trajectories end early, and the supplied RUL file gives the remaining cycles at the final observed cycle. All four raw subsets had zero missing values in the audited fields.

Subset	Train engines	Test engines	Train rows	Test rows	Fit windows	Features	Reg. / faults
FD001	100	100	20,631	13,096	14,459	18	1 / 1
FD002	260	259	53,759	33,991	37,051	24	6 / 1
FD003	100	100	24,720	16,596	17,754	19	1 / 2
FD004	249	248	61,249	41,214	42,960	24	6 / 2

Table 1. Official train counts precede the validation holdout. FD004 has 248 matched test engines and labels in the supplied files. The median run-to-failure history ranges from 199 to 234 cycles. NASA data repository

Preprocessing and leakage controls



Split. Seed 42 selects 20, 52, 20 and 50 validation engines for FD001-FD004. Training uses full windows with stride 1. Validation uses a fixed incomplete prefix per held-out engine. Test scoring uses the last observed window per official engine.

Features. Drop near-constant training sensors and retain the three operating settings. Standardize settings globally. Scale sensors globally for FD001/FD003 and within six training-fitted k-means regimes for FD002/FD004. Short windows repeat their first row on the left, with padding marked invalid.

Targets. Training RUL is final cycle minus current cycle. Fit to min(RUL, 125) / 125 and select checkpoints by capped validation RMSE. Multiply model output by 125 and retain it without clipping for raw and capped test scoring.

Accuracy and asymmetric risk

For e = prediction - target, MAE = mean(|e|), MSE = mean(e^2) and RMSE = √MSE. $R^2 = 1 - \Sigma e^2 / \Sigma(y - \bar{y})^2$. RAE (%) = 100 × Σ|e| / Σ|y - ȳ|. The C-MAPSS score sums exp(-e/13) - 1 for underestimates and exp(e/10) - 1 for overestimates, assigning a steeper penalty to late warnings. MAE, RMSE and bias are in cycles. MSE is in cycles². Higher R^2 and lower error or score are favorable.

Comparison unit. Main tables use raw official test labels. Report mean ± sample standard deviation over model seeds 42, 43 and 44 on the same engine split. These spreads reflect seed variability on a fixed test set.

03 BASELINE ARCHITECTURES

Twelve baseline designs

Each row traces the implemented forward path. Counts are FD001 inference parameters.

Mean 0 parameters	Training labels	→	Mean RUL	→	Constant RUL		
DeepNN 89,953 parameters	Masked flatten	→	ReLU 128 → 96 → 64 → 32	→	Linear RUL		
WideNN 138,753 parameters	Masked flatten	→	ReLU 256	→	Linear RUL		
RNN 3,313 parameters	Observed sequence	→	Tanh RNN 48	→	Final state	→	Linear RUL
LSTM 13,105 parameters	Observed sequence	→	LSTM 48	→	Final hidden	→	Linear RUL
GRU 9,841 parameters	Observed sequence	→	GRU 48	→	Final hidden	→	Linear RUL
BiLSTM 26,209 parameters	Observed sequence	→	LSTM 48 forward LSTM 48 backward	→	Concat final states	→	Linear RUL
CNN-LSTM 21,505 parameters	Masked sequence	→	Conv1D 48 × 3 ReLU, pad 1	→	LSTM 48	→	Final hidden → Linear RUL
Attention BiLSTM 30,914 parameters	Observed sequence	→	BiLSTM 48 + 48	→	96 → 48 → 1 masked attention	→	Weighted states → Linear RUL
Small LSTM + KD 2,321 parameters	Observed sequence	→	LSTM 16	→	Final hidden	→	Linear RUL
Fixed LIF SNN 3,265 parameters	Sensor currents	→	Recurrent LIF 48 leak .9 / threshold 1	→	Mean valid spikes	→	Linear RUL
ANN-to-SNN 89,953 parameters	Masked flatten	→	DeepNN weights calibrated IF layers	→	32-step mean final activity	→	Linear RUL

Figure 1. Recurrent blocks consume valid cycles. Small LSTM receives an extra teacher-output MSE during training. ANN-to-SNN uses training-calibrated thresholds and the DeepNN readout. The mean baseline predicts in cycles directly. Other scalar heads predict normalized RUL, which is rescaled for scoring.

Shared hyperparameters. 30 epochs, batch size 128, Adam at 0.001, gradient norm clipped to 1.0, window 30 and training stride 1. Models share the same protocol and epoch budget. Parameter counts vary with input width.

What the baselines test. DeepNN and WideNN test depth versus width with an ordered, flattened history. The plain RNN updates $h_t = \tanh(Wx_t + Uh_{t-1} + b)$. LSTM and GRU gate information through time. CNN-LSTM first extracts local patterns. The converted SNN tests whether a trained dense model's activity can be represented by firing rates over a short simulation.

04 PROPOSED METHOD

A spiking student with temporal guidance

One inference architecture supports four training variants and two single-loss removals.

FROZEN TEACHER / TRAINING ONLY



Training links ↓ Teacher hidden states → projected student membranes. Teacher attention → student attention. Teacher scalar → student scalar. All use MSE.

SPIKING STUDENT / DEPLOYED PATH



Figure 2. The teacher and the 48 → 96 training adapter (4,704 weights and biases) are absent from deployment. Student inference uses 3,409-3,697 parameters across subsets. The teacher uses 30,914-33,218.

Recurrence and loss

At each valid cycle t, candidate voltage $\tilde{v}_t = \lambda \odot v_{t-1} + W_i x_t + b_i + W_r s_{t-1}$. The neuron fires when $s_t = \mathbf{1}[\tilde{v}_t \geq \theta]$ and resets by subtraction, $v_t = \tilde{v}_t - \theta \odot s_t$. Leak λ = sigmoid(ℓ) and threshold θ = softplus(q) + 10^{-4} are learned per neuron, initialized at 0.9 and 1.0. Backpropagation uses the derivative surrogate $(1 + 10|\tilde{v}_t - \theta|)^{-2}$ and detaches the spike in the reset term. Padding keeps the recurrent state unchanged.

Attention α is the masked softmax of a linear score on membrane states. The scalar student readout is $\hat{y} = b + w^T \Sigma_t \alpha_t s_t$. The training objective is a weighted sum of label MSE, teacher-output MSE, projected membrane MSE, attention MSE and the mean spike rate over valid neuron-cycle slots. For the full student, the loss is $0.5L_{label} + 0.5L_{output} + 0.1L_{hidden} + 0.1L_{attention} + 0.01\rho$. Hidden and attention MSE use valid cycles only.

Training variant	Label	Output	Hidden	Attention	Spike rate
Learnable LIF SNN	1.0	0	0	0	0
Spiking output KD	0.5	0.5	0	0	0
Spiking temporal KD	0.5	0.5	0.1	0.1	0
SpikeDistill-RUL	**0.5**	**0.5**	**0.1**	**0.1**	**0.01**
Without hidden matching	0.5	0.5	0	0.1	0.01
Without attention matching	0.5	0.5	0.1	0	0.01

Training and evaluation pseudocode

1. Freeze the matched teacher at its best capped validation epoch.
2. For each training batch, compute teacher signals and unroll the student over valid cycles.
3. Add the configured losses. Backpropagate through the surrogate, clip gradients and update with Adam.
4. Retain the minimum capped validation RMSE checkpoint. Restore it and score raw and capped official test RUL.

05 OFFICIAL TEST RESULTS

Accuracy across models and subsets

Raw targets in engine cycles. The original Colab experiment contributes 192 model-subset-seed test runs.

Test RMSE by subset

Model	FD001	FD002	FD003	FD004
Mean	41.98 ± 0.00	53.78 ± 0.00	43.76 ± 0.00	54.56 ± 0.00
DeepNN	16.66 ± 0.46	**28.11 ± 0.55**	15.03 ± 0.58	27.84 ± 0.41
WideNN	16.98 ± 0.53	29.52 ± 1.38	16.77 ± 0.47	28.41 ± 1.28
RNN	16.21 ± 0.53	29.15 ± 0.70	16.79 ± 1.41	27.66 ± 0.52
LSTM	17.07 ± 0.81	28.90 ± 0.87	17.86 ± 1.21	26.70 ± 0.37
GRU	16.11 ± 0.50	29.23 ± 1.06	15.81 ± 0.57	**26.11 ± 0.40**
BiLSTM	15.75 ± 0.52	28.29 ± 0.36	15.32 ± 0.09	27.81 ± 0.61
CNN-LSTM	17.28 ± 0.73	30.59 ± 1.22	16.88 ± 0.60	27.67 ± 0.81
Attention BiLSTM	**15.72 ± 0.00**	29.05 ± 0.64	**14.70 ± 0.49**	27.10 ± 0.54
Small LSTM + KD	15.83 ± 0.44	29.39 ± 0.56	15.53 ± 1.48	26.75 ± 0.31
Fixed LIF SNN	19.66 ± 2.18	30.23 ± 0.70	21.69 ± 0.08	30.84 ± 0.93
ANN-to-SNN	44.18 ± 5.59	57.58 ± 8.30	31.65 ± 3.23	53.69 ± 6.30
Learnable LIF SNN	18.29 ± 0.71	29.44 ± 0.58	20.01 ± 1.71	29.27 ± 0.99
Spiking output KD	17.37 ± 0.36	29.06 ± 0.74	20.32 ± 1.82	29.21 ± 0.42
Spiking temporal KD	17.07 ± 0.98	29.65 ± 0.03	19.07 ± 1.37	28.47 ± 0.87
SpikeDistill-RUL	**17.67 ± 0.11**	**29.00 ± 0.06**	**20.18 ± 0.51**	**28.33 ± 0.79**

Table 3. Mean ± sample SD over seeds 42-44. Teal numbers identify the lowest mean in each subset.

Cross-subset summary of the other test criteria

Model	MAE	MSE	RAE (%)	R^2	Score / engine
Mean	41.21	2,386.7	100.39	-0.035	415.05
DeepNN	15.45	517.4	37.08	0.793	28.38
WideNN	16.64	562.8	40.05	0.774	44.34
RNN	16.12	540.4	38.74	0.783	26.61
LSTM	16.07	540.0	38.73	0.779	23.66
GRU	15.49	511.7	37.22	0.795	21.61
BiLSTM	15.74	514.3	37.75	0.796	19.41
CNN-LSTM	16.24	571.6	38.97	0.770	33.31
Attention BiLSTM	15.36	510.6	36.76	0.798	23.22
Small LSTM + KD	15.32	518.3	36.74	0.793	24.88
Fixed LIF SNN	19.04	681.5	46.08	0.716	40.89
ANN-to-SNN	36.12	2,312.7	87.49	0.032	1,285.35
Learnable LIF SNN	17.80	615.2	43.05	0.746	33.29
Spiking output KD	17.57	603.7	42.48	0.751	24.96
Spiking temporal KD	16.83	586.8	40.58	0.760	31.88
SpikeDistill-RUL	**17.47**	**590.9**	**42.28**	**0.755**	**25.93**

Table 4. Each subset receives equal weight. Score is divided by its test engine count before averaging. Complete subset-level raw and capped metric tables · per-seed scores

Metric choice changes the ranking. On FD001 the teacher has the lowest mean RMSE (15.72), the small distilled LSTM has the lowest MAE (11.80), and BiLSTM has the lowest asymmetric score (357.77). On FD003, the teacher leads RMSE while DeepNN has the lowest score (323.81).

06 ABLATIONS AND TRAINING

What the transfer losses contribute

A separate CPU experiment retrained all three arms under the same conditions.

Within each subset and seed, all three runs shared the frozen teacher, 48-neuron student, optimizer, training snapshot and 30-epoch budget. Each removal sets just one coefficient to zero. Attention pooling and spike regularization stay active. The full-model control below belongs to this rerun.

Subset	Full control	No hidden loss	No attention loss	Δ hidden	Δ attention
FD001	17.07 ± 1.13	17.46 ± 0.85	17.26 ± 1.18	+0.39	+0.19
FD002	29.56 ± 0.09	29.91 ± 0.88	30.18 ± 0.37	+0.35	+0.61
FD003	19.88 ± 0.46	21.26 ± 0.87	20.48 ± 0.97	+1.38	+0.60
FD004	28.81 ± 0.20	28.84 ± 1.03	29.09 ± 0.57	+0.04	+0.28

Table 5. Raw test RMSE in cycles, mean ± sample SD across three matched seeds. Δ is the mean within-seed increase after removing a loss. All 36 runs were scored on their official engine sets.

FD003 has the clearest hidden-state contribution (+1.38 cycles). The FD004 hidden-state change is only +0.04 cycles. The FD004 asymmetric score favors the attention removal (6,774 vs 7,584 for the full control). These are descriptive three-seed differences. The original Colab sparsity contrast is also mixed: adding the rate term improved RMSE on FD002/FD004 and increased it on FD001/FD003.

Validation curves from the matched ablation runs

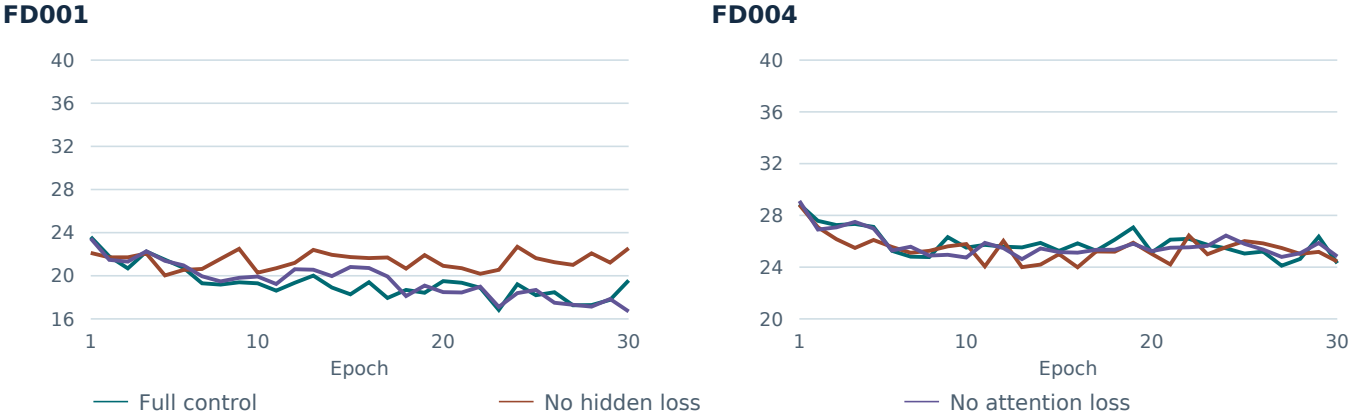


Figure 3. Capped validation RMSE (cycles) by epoch, averaged over seeds 42-44. FD001 and FD004 illustrate the lower- and higher-error validation trajectories. Checkpoint selection is performed independently per seed.

Observed spike activity on official test windows

Subset	Mean valid steps	Spike rate	Recurrent SynOps / window	Student params
FD001	30.00	23.74%	15,834	3,409
FD002	29.88	20.73%	13,749	3,697
FD003	30.00	22.42%	14,938	3,457
FD004	29.72	19.67%	12,990	3,697

Table 6. Original full student, seed 42. The model unrolls 30 steps. Spike rate is events / valid neuron-step slots. Estimated recurrent SynOps = 48 × preceding spikes that reach a valid next step. Input/readout operations and dense kernel overhead are outside this estimate. Ablation and figure bundle

For the ablations, the prepared archive bytes differed from the Colab fingerprints. Matched teacher validation endpoints, labels and predictions were verified on the snapshot used by all three arms.

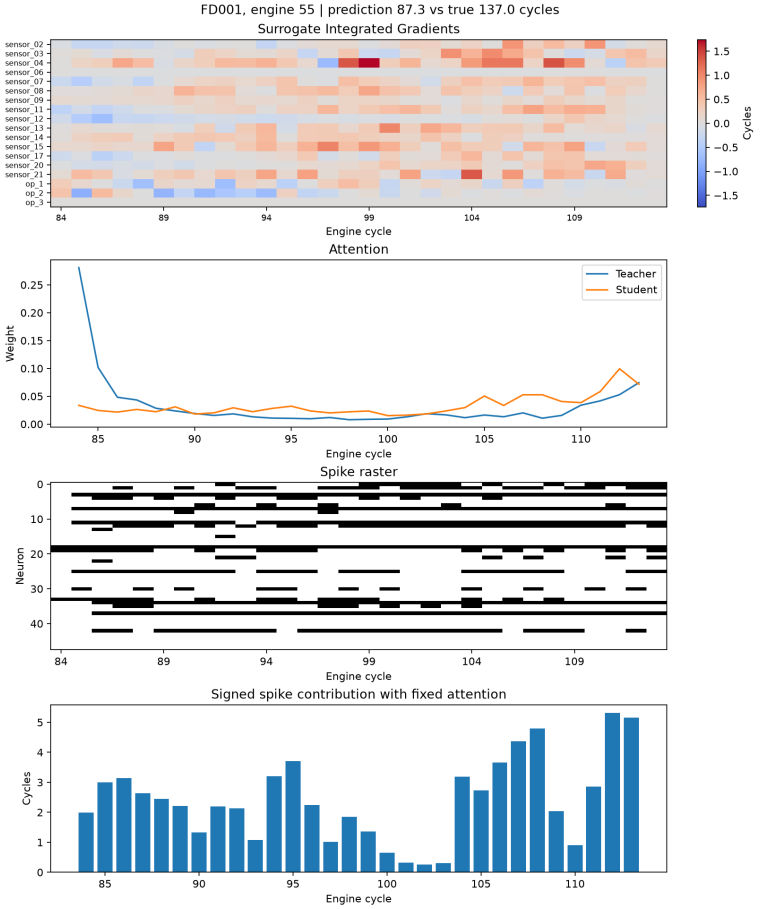
07 EXPLAINABILITY AND ERROR ANALYSIS

How the student uses the window

Original saved SpikeDistill-RUL checkpoint, seed 42, with matched teacher outputs.

Subset	Lowest-error engine	Pred. / actual	Highest-error engine	Pred. / actual
FD001	49	20.83 / 21	55	87.27 / 137
FD002	252	12.03 / 12	86	96.20 / 194
FD003	52	120.23 / 120	16	121.55 / 56
FD004	151	31.07 / 31	54	100.20 / 176

Table 7. Lowest and highest absolute raw RUL errors in each official test subset. Cases are selected for diagnosis.

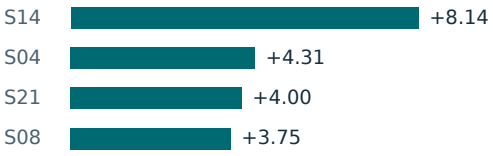


Temporal SHAP · engine 55

	84-88	89-93	94-98	99-103	104-108	109-113
S07	-5.8	-4.5	-2.8	-1.0	+3.5	+3.1
S02	-3.9	-0.8	-7.8	+0.0	+1.2	+3.8
S08	-1.9	+2.7	+1.6	+2.3	-0.0	+3.3
S04	+0.3	+1.5	+0.0	+2.1	+3.6	+3.9
S11	-2.6	-0.8	-0.5	+0.5	+2.0	+3.5
S12	-4.1	-2.6	-1.9	-0.4	-0.9	-0.0
S17	-4.0	-0.9	-2.3	+0.4	+0.9	+1.0
S20	-2.0	-0.8	-1.1	-1.6	-0.5	+2.9

Signed cycles vs zero-sensor reference. Eight sensors shown by total absolute contribution.

FD001 sensor permutation



Change in raw RMSE (cycles), mean of five shuffles.

Engine 55. The teacher emphasizes the start of the window. The student shifts more weight toward later cycles and underestimates RUL by 49.73 cycles.

Figure 4. FD001 high-error example and subset-level permutation summary. Full-resolution figures include every subset. Download the 20 figures

Temporal SHAP and permutation. Shapley sampling uses 32 orderings of five-cycle sensor blocks. Permutation shuffles a whole sensor trace among engines with equal valid-window length. Both measure sensitivity to their chosen perturbations.

Surrogate IG, attention and spike attribution. IG uses 64 path points and has large completeness residuals under hard spikes. Attention shows temporal weighting. At fixed attention, the signed spike contributions plus bias recover the scalar readout.

Failure modes. FD002 engine 86 has 194 cycles left and is underestimated by 97.80 cycles. The 125-cycle training cap provides context. FD003 engine 16 is overestimated by 65.55 cycles, the direction the asymmetric score penalizes more steeply.

08 DEPLOYMENT AND CONCLUSION

Deployment status and references

The validation-selected FD001 student (seed 42, epoch 14) was exported as an FP32 ONNX opset-17 graph with batch size one. It accepts a [1, 30, 18] standardized feature tensor and a [1, 30] validity mask, and returns RUL directly in cycles.

Check	Verified result	Scope
ONNX prediction parity	Max |Δ| = 0.000046 cycles	100 FD001 engines + 1 padded case
ONNX raw test RMSE	17.55 cycles	FD001, seed 42
ONNX latency	0.720 ms median / 1.226 ms p95	Windows CPU, batch 1
Jetson latency, memory, power, energy	Pending measurement	Physical Orin Nano

Table 8. CPU timing used ONNX Runtime, 100 warm-ups and 1,000 calls. The target-board protocol uses ≥1,000 timed calls after warm-up and records power mode, precision, peak memory, rail power and energy per inference. FP32 TensorRT build, parity and timing remain pending on the board. ONNX bundle · Jetson benchmark package

Conclusion and limitations

The full student is compact and improves on fixed LIF regression. Temporal and attention guidance have subset-dependent effects, with the strongest isolated change on FD003. Attention BiLSTM leads FD001/FD003, DeepNN leads FD002 and GRU leads FD004 by mean raw RMSE. Practical deployment needs physical power and latency measurements. Three seeds, simulated data, capped training targets and local attribution choices limit generalization. Future experiments can test uncapped or risk-weighted training and validation on field engines.

Reproducibility. Code and notebooks · original baseline runs · original student runs · controlled runs and explanations

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